

Simulating a hanging cable: mathematics, measurement, and five solvers compared

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Abstract

A flexible cable is suspended from 3 level clamps spanning $S = 0.350$ m and photographed. Its shape is predicted three independent ways: from the governing differential equation, from a set of physics solvers, and from the photograph itself. The three-way structure is deliberate — any two of them can agree for uninteresting reasons, and only the third makes it possible to attribute a disagreement to numerical error rather than to modelling error. This document reports what agrees, what does not, and which results are currently trustworthy.

1 Apparatus

The cable is clamped with tape to a vertical door at 0.000 m, 0.150 m, 0.350 m measured from the leftmost clamp, all at one height, and photographed square-on. Cable length is not taken from the packaging: the free length between clamps is what governs the shape, so it is measured from the same photograph the simulations are compared against. Here that gives $L = 0.8486$ m, and $L/S = 2.425$ — a very slack cable, sagging further than it spans.

2 Image processing

The cable is separated from the door in HSV: it is bright and nearly unsaturated where the door is a saturated brown. The connected component is then selected by SHAPE rather than by area — a specular highlight on the door is also bright and unsaturated, and is often *larger* than the cable, so selecting the largest region finds the glare instead. A cable is thin and spans the clamps; glare is a blob.

The centreline is then traced ALONG the curve: from each point the tracer advances one step along the local tangent and re-centres on the mask's perpendicular cross-section. The obvious alternative — averaging the mask rows in each pixel column — fails badly here, because a slack cable hangs near-vertically close to its clamps and one column then spans a long run of cable. Figure 2 is the check on all of this: the left panel tints every segmented pixel so a mis-segmentation is visible directly, and the right panel compares the extracted profile to a fitted catenary, because a clean segmentation can still fit badly and those are different failures.

3 Mathematical model

For a perfectly flexible, inextensible cable of weight w per unit length, the horizontal component of tension is constant along the cable, $T \cos \theta = H$. Balancing the vertical forces on an element ds and substituting $ds = \sqrt{1 + z'^2} dx$ gives the governing equation

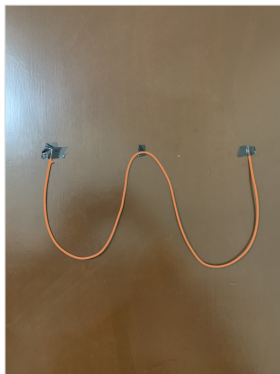
$$z'' = \frac{\sqrt{1 + z'^2}}{a}, \quad a = \frac{H}{w}, \quad (1)$$



Figure 1: The cable as photographed. The tape fixes the cable's ANGLE as well as its position, which is a clamped boundary condition, not the pinned one the classical catenary assumes.

Cable detection: three_point.jpg

1. Detected cable pixels



135,298 pixels segmented (HSV: low saturation, high value)

2. Profile vs catenary (RMS residual 67.9 mm)

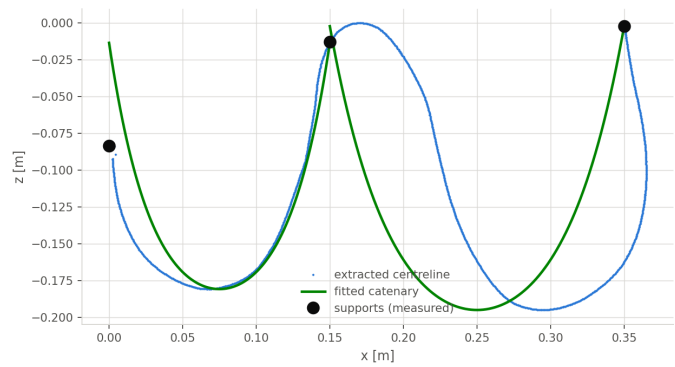


Figure 2: Detection and fit. Left: every segmented pixel, tinted over the original. Right: extracted profile against the fitted catenary.

a two-point boundary-value problem closed by the arc-length constraint $\int_0^S \sqrt{1 + z'^2} dx = L$, which is what fixes a . Physically a is set by how much cable was hung: a slacker cable has a smaller a and therefore carries *less* tension.

The corresponding dynamics, which is what the solvers actually integrate, is

$$\rho A \frac{\partial^2 \mathbf{r}}{\partial t^2} = \frac{\partial}{\partial s} \left(T \frac{\partial \mathbf{r}}{\partial s} \right) + \rho A \mathbf{g}, \quad (2)$$

with $|\partial \mathbf{r} / \partial s| = 1$ enforced by T , which is a Lagrange multiplier rather than a material property. Setting $\partial^2 \mathbf{r} / \partial t^2 = 0$ recovers (1), which is the assumption that licenses scoring a settled simulation against the static solution.

Verification. Equation (1) is integrated numerically (RK4 from the vertex, bisection on a for the length constraint) without using the closed form $z = a \cosh((x - x_0)/a) + c$ anywhere. The two agree to $7 \times 10^{-3} \mu\text{m}$. The analytic reference every error in this document is measured against is therefore verified rather than assumed.

4 Simulation

Each solver runs in its own process — Isaac Sim’s `SimulationApp` is a singleton and the methods force different physics engines — and each is handed the identical scenario file, so a method that disagrees is disagreeing about physics rather than about its inputs.

method	RMSE [mm]	max [mm]	sag [mm]	arc drift [%]	settled	wall [s]
Newton cable (add_rod + VBD)	1.49	14.17	207.5	2.813	yes	15.2
PhysX capsule chain (D6)	73.10	153.90	186.8	0.024	NO	15.8
Warp XPBD rod	159.53	295.24	189.1	-0.461	yes	0.6

Table 1: Each solver against the analytic catenary. Arc drift is the reference-free check: an inextensible cable must keep its length whatever the catenary says.

Boundary conditions are not a detail. These methods do not all hold the middle clamp the same way. The Newton cable method clamps it, fixing the cable’s angle as the tape does; the others pin it and let the cable pivot. That difference alone accounts for a large part of the spread above, so the table compares *models*, not solver quality. The middle clamp was measured to hold 45.7% of the cable’s arc length on its left, where the chord-proportional convention assumes 42.9% (+2.8 percentage points). Since the clamp pins a material point, this split is a property of the apparatus rather than something statics determines, so it must be measured and imposed.

method	RMSE vs photo [mm]	max [mm]
Newton cable (add_rod + VBD)	86.64	191.76
PhysX capsule chain (D6)	36.10	98.45
Warp XPBD rod	175.61	278.57

Table 2: Simulation against the photographed cable. PROVISIONAL — see Section 7; the extraction’s frame is known to be skewed, so these numbers are reported for completeness and must not be quoted as results.

5 Solver hyperparameters

Every cable example shipped with Newton uses 10 substeps and 5 iterations. Applied to this problem that is far too coarse; but the repository’s previous default of 8×200 was tuned by

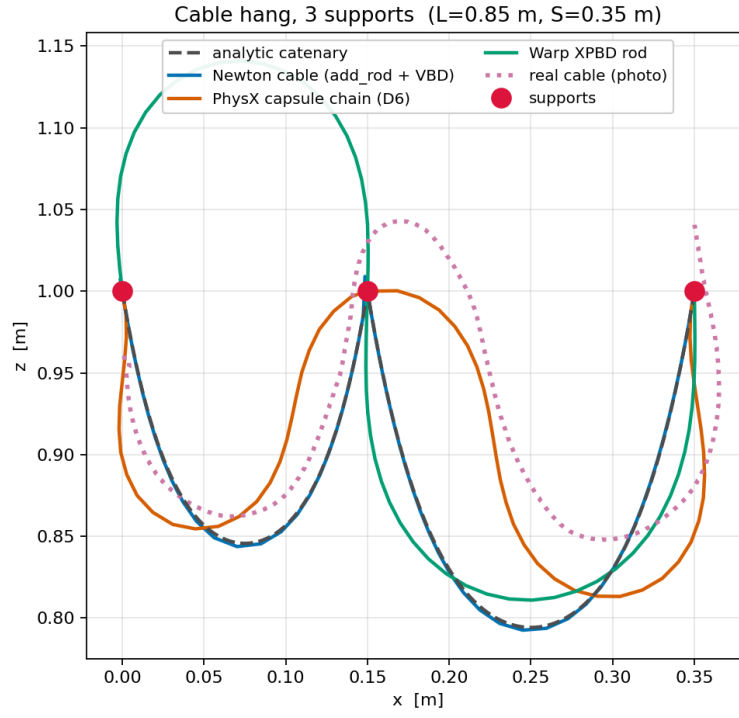


Figure 3: All methods, the analytic catenary, and the photographed cable on one axis.

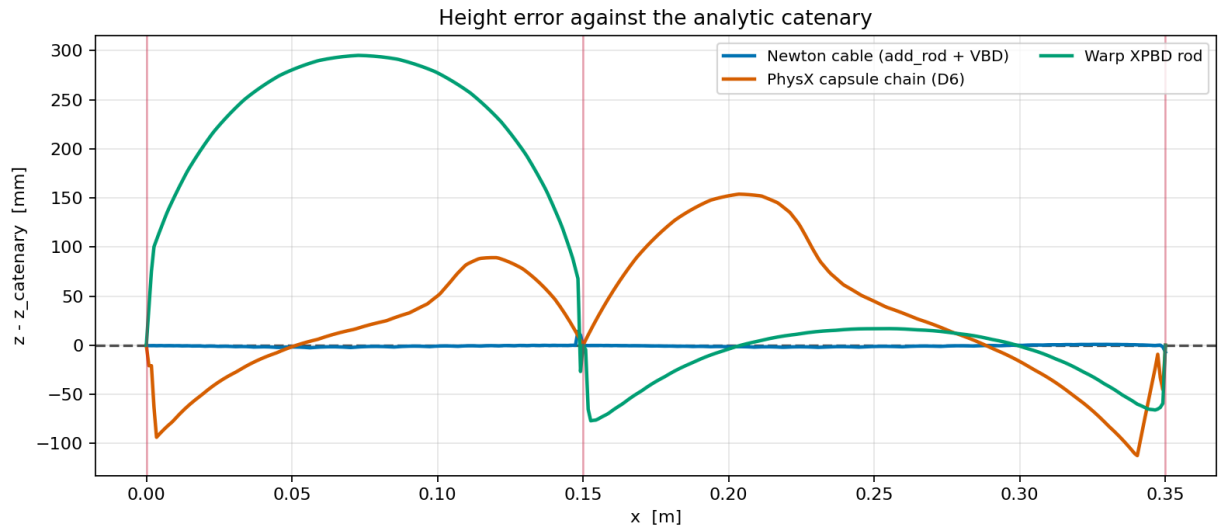


Figure 4: Deviation from the analytic catenary along the cable.

varying iterations alone, leaving the substep axis untested. A sweep over both, scored against the analytic curve, shows accuracy to be essentially a function of the *product* — while cost is not, because each substep carries fixed overhead that an iteration does not. Fewer substeps with more iterations is therefore strictly cheaper at equal accuracy.

Separately, damping the stretch constraint proved actively harmful: it opposes the solver’s own length correction, and raising it to 10^{-2} cost roughly a factor of six in accuracy for no benefit. Damping the *bending* constraint, by contrast, changed nothing measurable over a $600\times$ range, because this cable’s *EI* is small enough that the bending mode carries almost no energy.

substeps	iterations	budget	RMSE [mm]	arc drift [%]	wall [s]
4	5	20	70.93	14.51	6.4
10	5	50	27.81	5.36	10.1
4	200	800	2.56	0.44	65.7
8	200	1600	1.53	0.22	190.1
10	200	2000	1.31	0.17	239.6
16	200	3200	2.09	0.12	379.7

Table 3: Selected configurations from the sweep.

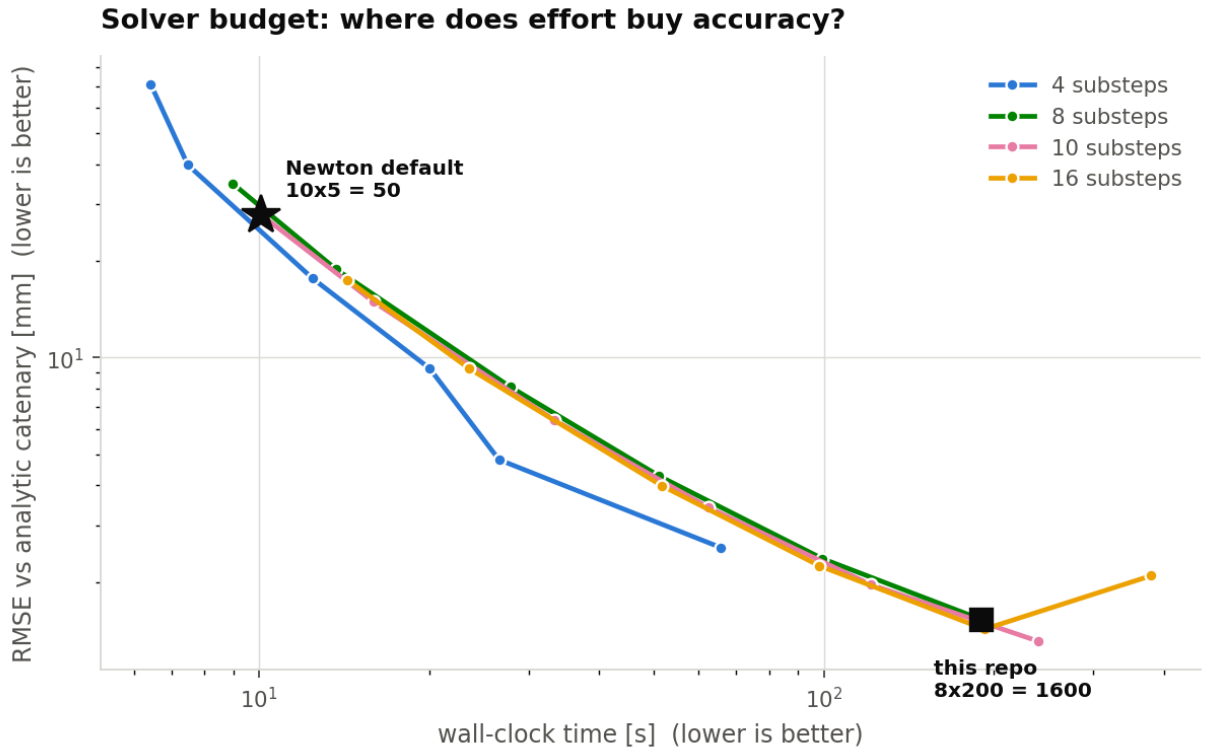


Figure 5: Accuracy against wall-clock cost. Both axes logarithmic; the knee is where further cost stops buying accuracy.

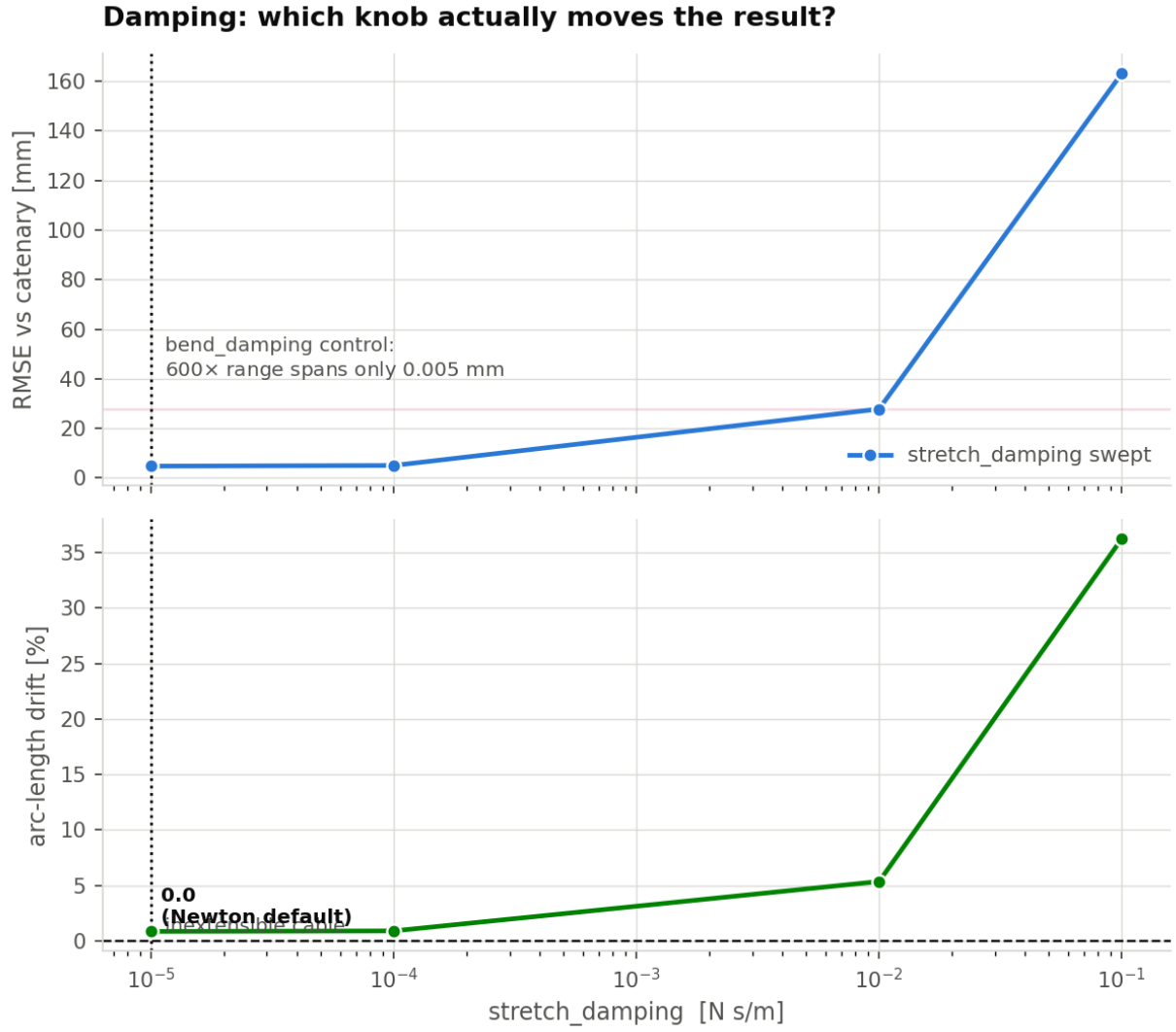


Figure 6: Damping. Stretch damping dominates; the bend-damping control spans a 600x range and does nothing.

field	source	value
<code>cable.length_m</code>	measured	0.8486
<code>supports.mid_fraction</code>	measured	0.4570

Table 4: Which inputs were measured from the photograph and which were assumed. Recorded automatically for every run.

6 Provenance of the inputs

7 Validity

Simulation-versus-photograph comparison is NOT yet valid

The extraction’s coordinate frame is skewed. The two outer clamps are physically level, but come out 81 mm apart in height, and the middle clamp’s peak appears at $x = 0.172$ m where it was measured at 0.150 m.

The cause is identified and is in the extraction, not the apparatus: the tracer’s two endpoints were defined by different rules. One was the mask’s horizontal extreme, the other the point where tracing halted at the tape. Those coincide for a taut cable, but this cable is slack enough that the clamped end bulges horizontally *past* its own anchor, so the two rules pick different points and the frame tilts.

Consequence. Every quantity derived from the photograph’s absolute frame — in particular any simulation-versus-photograph RMSE — is provisional and is therefore *not* reported as a finding here. The fix is to define both anchors identically, by detecting the tape and taking the point where the cable enters it. Recovering the middle clamp at 0.150 m would be an independent confirmation that it worked.

Unaffected by the above, because no photograph enters them:

- the verification of (1) against its closed form;
- every solver’s error against the analytic catenary;
- the hyperparameter sweep and its conclusions;
- the observation that the Warp rod diverges at this slackness, rising above the clamps rather than hanging between them.

Modelling limits. The catenary is the $EI \rightarrow 0$ limit. The real cable is a charger cable with a permanent set, so its rest shape is not straight and some deviation from any ideal curve is physical rather than numerical. The tape also clamps the cable’s angle, which the classical catenary does not model. Both effects are concentrated near the clamps, and both are measurable once the frame is corrected — which makes them results to quantify rather than caveats to apologise for.